

Claude Multi-Account Fine Tuning

Changelog

v1.12.2 — 2026-07-05 — Native alias auto-registration + account-detection hardening

Fixed

Added

v1.12.1 — 2026-07-05 — Judge independence + resolve/robustness hardening

Changed

Added

Fixed

v1.12.0 — 2026-07-05 — Semantic code-visibility (layer 3) + live TUI verification (layer 4)

Added

Fixed

Notes

v1.11.2 — 2026-07-03 — Token-limit guard + re-enabled provider proxies + Poe tool cap

Fixed

Changed

Added

v1.11.1 — 2026-07-03 — Live per-alias Claude Code verification (CLI + TUI) + provider fixes

Added

Fixed

Verified (live on host)

v1.10.8 — 2026-07-01 — noclobber-safe router-provider config write

Fixed

Added (tests)

Verified

Note (not a toolkit bug)

v1.10.7 — 2026-06-29 — Shared-items drift guard + audit closure

Added (tests)

Audited — no code changes (3 parallel investigators; every finding independently checked)

Verified

v1.10.6 — 2026-06-29 — Committed credential-leak regression test
Added (tests)
Verified

v1.10.5 — 2026-06-29 — Provider 'null' field normalization + coverage
Fixed
Added (tests)
Audited — no change needed (independently verified, not taken on trust)
Verified

v1.10.4 — 2026-06-29 — set -e/pipefail abort fixes + hardened test coverage
Fixed
Changed (tests)
Verified

v1.10.3 — 2026-06-29 — Execution-level wrapper test coverage
Added
Verified

v1.10.2 — 2026-06-29 — Self-healing rc source lines + strict rc tests
Fixed
Added
Verified

v1.10.1 — 2026-06-29 — Robust cma_run wrapper assertions
Fixed
Verified

v1.10.0 — 2026-06-29 — Auto-applied per-alias session color + coverage/wiring
Added
Fixed
Verified

v1.9.2 — 2026-06-29 — Hermetic CLAUDE_BIN resolver test
Fixed
Verified

v1.9.1 — 2026-06-29 — CLAUDE_BIN resolves across per-host install locations
Fixed
Verified

v1.9.0 — 2026-06-29 — Per-project auto-sessions that actually work live + zero-coverage tests
Fixed
Added
Verified

v1.8.1 — 2026-06-29 — Merge-engine correctness + portability hardening
Fixed
Added
Verified

v1.8.0 — 2026-06-29 — Alias isolation + token-limit guard + per-project auto-sessions
Fixed

Added
Verified
v1.7.12 — 2026-06-28 — One-line curl installer
Added
Verified
v1.7.11 — 2026-06-28 — Round-4: coverage-gap regression tests, toon recursion guard, arg validation
Fixed
Added — coverage-gap regression tests (the same class that let enable-plugins
Verified
v1.7.10 — 2026-06-28 — Round-3 audit: enable-plugins bug fix, path-traversal guards, proxy robustness
Fixed
Security (defense-in-depth)
Robustness
Verified
Audit (round 3) — verified clean
v1.7.9 — 2026-06-28 — Hardening round 2: injection-safe alias writes, broadened secret redaction, docs accuracy, shellcheck 0
Security
Fixed
Docs
Quality
Verified
v1.7.8 — 2026-06-28 — Secret hygiene (argv + committed-proof leaks), dead-code fix, coverage tests
Security
Fixed
Added
Verified
v1.7.7 — 2026-06-28 — Portable realpath (BSD portability hardening), set -u edge fix, regression tests
Fixed
Added
Verified
Audit findings (v1.7.6 — no code change required)
v1.7.6 — 2026-06-28 — Always-non-interactive execution, alias-file integrity, macOS/bash-3.2 portability, 4-host rollout
Fixed
Added
Multi-host rollout (nezha · mistborn.local · thinker.local · amber.local)
Verified
v1.7.5 — 2026-06-28 — Cross-provider /resume session visibility fix
Fixed
Root Cause Analysis
Verified
Tests

v1.7.4 — 2026-06-26 — Kimi provider fix + AWS IaC MCP disabled by default
Fixed
Changed
Tests

v1.6.6 — 2026-06-21 — TOON integration for token-efficient prompts
Added
Token Savings
Note
Tests

v1.6.5 — 2026-06-21 — Poe proxy fix (alias file + install)
Fixed
Verified
Tests

v1.6.4 — 2026-06-21 — Poe proxy fix for tool compatibility
Fixed
Verified
Tests

v1.6.3 — 2026-06-21 — Poe provider (382 models, 3 aliases)
Added
Verified

v1.6.2 — 2026-06-21 — Chutes provider documentation + model update
Changed
Verified

v1.6.1 — 2026-06-21 — cache_control fix + E2E tests
Fixed
Added
Verified working (all aliases tested with "Do you see our codebase?")

v1.6.0 — 2026-06-21 — Multi-alias provider system
Added
Changed
Full test suite
Usage

v1.5.1 — 2026-06-20 — Linux stat fix + nezha deployment
Fixed
Deployment
Full test suite

v1.5.0 — 2026-06-20 — Cross-alias session visibility
Added
Changed
Full test suite
How it works
Performance

v1.4.0 — 2026-06-20 — OpenCode Zen provider alias
Added
Changed
Full test suite
Honest notes

v1.3.0 — 2026-06-19 — Xiaomi MiMo provider alias
Added
Changed
Full test suite
Honest notes

v1.2.0 — 2026-06-19 — Z.AI Coding Plan provider alias
Added
Changed
Full test suite

v1.1.0 — 2026-06-16 — Distributed infrastructure + provider verification
Added
Changed
Fixed (LLMsVerifier — shipped as PR #2, applied to deployed builds)
Verified live (real "Do you see my code?" against production APIs)
Safety

v1.0.0 — 2026-06-16 — Dynamic provider-alias generator

v1.6.7 — 2026-06-21 — Poe proxy fix for all aliases
Fixed
Verified
Tests

v1.6.8 — 2026-06-21 — Poe proxy gzip fix
Fixed
Verified
Tests

v1.6.9 — 2026-06-21 — Poe proxy \$ref fix for Grok-4
Fixed
Verified
Tests

v1.7.0 — 2026-06-22 — Poe proxy complete fix (all aliases verified)
Fixed
Verified (all three aliases through full Claude Code flow)
Root Cause Analysis
Tests

v1.7.1 — 2026-06-22 — Full validation + release
Fixed
Tests
Release

v1.7.2 — 2026-06-22 — Claude alias verification, full release
Added
Tests

Changelog

All notable changes to the Claude multi-account toolkit.

v1.12.2 — 2026-07-05 — Native alias auto-registration + account-detection hardening

Fixed

- **Native `claude<N>` aliases were never created for pre-existing account dirs.** Running `install.sh` or `claude-unify.sh` only merged shared state; if account directories already existed (e.g. `~/.claude-milos85vasic`), no shell aliases were registered, so `claude1/claude2/etc.` were undefined. `claude-unify.sh` now auto-registers a `claude<N>` alias for every detected account that lacks one.
- **Smart alias numbering:** an account dir literally named `~/.claude-claude4` keeps the `claude4` alias; remaining dirs fill the lowest free `claude<N>` slot instead of skipping over reserved numbers.
- **Bogus account detection:** `~/.claude-code-router` and `~/.claude-*.lock` directories are no longer treated as Claude accounts, preventing them from stealing alias slots or being merged into shared state.

Added

- Regression tests in `test_unify.sh` and `test_install.sh` prove that `install/unify` register native aliases for existing account dirs and preserve `claude<N>` basenames while filling gaps.

v1.12.1 — 2026-07-05 — Judge independence + resolve/robustness hardening

Addresses the v1.12.0 final whole-branch review's deferred items and the deep-research findings on LLM-as-judge bias.

Changed

- **Round-2 judge default is now a DIFFERENT model family.** `providers/judge.env.template` defaults to Groq / Llama-3.1 (`llama-3.1-8b-instant`) instead of DeepSeek. 2024-2026 research (arXiv:2508.06709 and others) shows a judge systematically favors its own model family — including validating that family's *wrong* answers, the exact failure layer-3 exists to catch — so defaulting the judge to a common subject (DeepSeek) was the worst case. Verified working live as an independent judge.

- **The semantic command distinguishes transport/infra failures from genuine verdicts.** The LLMsVerifier semantic-code-visibility command now exits **3** when a round-1/round-2 API call cannot complete (non-2xx, timeout, empty, connection error), vs exit **1** for a real negative verdict. providers-semantic.sh maps exit 3 → skip, so a transient judge/model hiccup is an honest SKIP and never downgrades the model-under-test (final-review I-1).

Added

- **Independence warning:** providers-semantic.sh warns (never fails) when the judge endpoint equals the model-under-test endpoint (same provider = same family = not independent).
- **xAI is now resolvable:** providers/overrides.json gains xai → <https://api.x.ai/v1> (the catalog lists xAI with no API base, so it previously resolved unmapped). Endpoint confirmed live.
- CONST-069 capability-boundary mandate in the LLMsVerifier submodule constitution (records the CONST-051 project-agnostic boundary under a non-colliding id).

Fixed

- A directory passed as the keys file (CMA_KEYS_FILE / --keys-file) now dies with a clear message instead of silently yielding "0 key vars".

v1.12.0 — 2026-07-05 — Semantic code-visibility (layer 3) + live TUI verification (layer 4)

Adds two new provider-verification layers on top of the Phase-1 existence/tool-call checks, driven by the LLMsVerifier submodule's standalone, stdlib-only semantic-code-visibility Go command. The whole pipeline was proven end-to-end against real providers (redacted, real network): a genuinely code-seeing model (mistral-medium-2604) verifies (round-1 sentinel + round-2 judge 2/3); models that bluff or fail round-2 (chutes GLM-5.2-TEE — empty/timeout) are unverified; billing blocks (deepseek 402) are unverified — never a faked pass.

Added

- **Layer 3 — semantic code-visibility.** scripts/providers-semantic.sh renders the toolkit-owned rubric (providers/rubric/code-visibility-rubric.json) into a judge prompt and drives the Go command through a build-and-cache driver (scripts/claude-

semantic-visibility.sh). Two rounds: a unique sentinel embedded in a code fixture (does the model actually see the code?), then an independent LLM-as-judge score ($\geq$ threshold). One-word contract verified|unverified|skip (exit 0/1/2). Wired into cmd_sync; keys move via env only (never argv). CONST-051 boundary held — the submodule stays project-agnostic, receiving fixture/prompt/ judge-prompt/sentinel only as CLI args.

- **Layer 4 — live superpowers-TUI verification.** scripts/verify_superpowers_tui.sh launches real Claude Code through a provider alias and confirms the superpowers plugin engages with no trust/overwrite prompt — the only thing that flips a provider to fully verified. Honest SKIP when preconditions (real claude/key/network) are absent; the engagement classifier is hardened against false-PASS on prompt-echo, and a live negative-case test (neutral prompt → marker must NOT match) proved it does not false-verify.
- **claude-providers verify <id> [--deep]** — single-provider deep re-verify.
- **Tier-B live verifier** (scripts/tests/verify_providers_live.sh) runs layers 3–4 per installed provider, writes secret-redacted evidence + an aggregate proof/providers-summary.json; already wired into run-proof.sh.

Fixed

- **--refresh-aliases was not byte-idempotent** — cma_provider_write_alias re-ran the full cma_ensure_alias_file self-heal migrations on every alias line, which could non-deterministically reposition the cma_run_provider function and produce a different file on a second refresh (the session hook runs this on every shell). Now only bootstraps the alias file when absent. (Root-caused from a captured diff; 42 flake-loops + repeated full-suite runs now deterministic.)
- Three bugs found by real execution of the new layer-3 path: claude-semantic-visibility.sh --help failed pre-build; providers-semantic.sh discarded the driver's JSON evidence to /dev/null; the judge base URL was not normalized (a trailing /v1 doubled to /v1/v1/chat/completions).
- Final-review hardening: cmd_sync/cmd_sync_multi heal a stale/outdated cma_run_provider wrapper once per sync (the byte-idempotence change had removed the per-shell self-heal, so a pre-Phase-2 gate-less wrapper could still launch); the layer-4 engagement classifier now requires the namespaced superpowers:<name> form (a bare skill word could false-verify); and verify --deep treats a layer-4 verifier crash as an honest SKIP rather than a clean verify.

Notes

- The default providers/judge.env.template judge is DeepSeek. For strongest results, configure providers/judge.env with a judge from a **different model family** than the provider under test — 2024-2026 research (e.g. arXiv:2508.06709) shows same-family judges systematically favor their own family's outputs, including validating wrong answers. A different-family default, an xAI overrides.json base-url entry, and the submodule boundary-contract constitution id are tracked for a follow-up.

v1.11.2 — 2026-07-03 — Token-limit guard + re-enabled provider proxies + Poe tool cap

Ran every provider alias through REAL Claude Code executing / using-superpowers (the user's reproducing prompt), in CLI mode with scrubbed env, and root-caused every genuine failure. Three real toolkit bugs fixed (below); all remaining non-passes are account-side (insufficient funds, rejected/absent keys, or a key with no chat entitlement) — not toolkit defects.

Fixed

- **Input token-limit 400** — scripts/lib.sh (cma_run_provider) now exports
CLAUDE_CODE_AUTO_COMPACT_WINDOW="\$CMA_PROVIDER_CONTEXT_LIMIT"
before the transport branch, so Claude Code knows each provider's real input window and auto-compacts (at window - 13000) before a request overshoots it. Fixes the user-reported 400 ... exceeded model token limit: 262144 (requested: 311786) on kimi-for-coding and any provider whose window is smaller than the ~1M Claude Code assumes for Anthropic's endpoint. Fully dynamic/parametrized: the value is the per-model limit.context from the models.dev catalog, persisted in each provider .env. Guarded by [[-n ...]]; applies to **both** transports. (CLAUDE_CODE_MAX_OUTPUT_TOKENS only ever capped **output** — it could not fix an **input** overflow.) Verified live: kimi's /using-superpowers now compacts and succeeds.
- **All provider proxies were silently disabled** — cma_run_provider resolved compatibility proxies via \$LIB_DIR/proxy/..., but LIB_DIR is a repo-only variable that does not exist in the self-contained alias file, so it expanded empty and **no proxy ever started** (Poe, etc.). Now resolves against the installed location \${SHARED_DIR:-\$HOME/.claude-shared}/proxy (where install.sh copies scripts/proxy/*.py). This is why Poe returned 400 Invalid 'tools': Field required — its request

never went through the proxy that injects the required parameters field.

- **Poe tool-count limit** — Poe rejects requests carrying more than ~216 tool definitions with the same misleading 400 Invalid 'tools': Field required (verified count-based: 215 accepted, 220 rejected, independent of payload size). On this host Claude Code's large MCP-plugin load emits 400+ tools. `scripts/proxy/poe_proxy.py` now caps the tool list to `P0E_MAX_TOOLS` (default 200), dropping only overflow mcp_... tools so **every** built-in Claude Code tool is preserved. Parametrized via the `P0E_MAX_TOOLS` env var. Verified live: Poe's `/using-superpowers` now returns a successful result.

Changed

- **cma_ensure_alias_file migration** — added `CLAUDE_CODE_AUTO_COMPACT_WINDOW` and `_cma_proxy_dir` as regeneration markers, so an already-installed `cma_run_provider` predating either fix is transparently regenerated on the next alias-file touch.
- **scripts/tests/verify_claude_live.sh** — `reclassify_fail` now maps a direct 400 whose body says the model is unknown/invalid to **BADKEY** (an account that can list models but not invoke them — e.g. inference here), while a 400 with no model-rejection marker (a real launch-layer defect) stays FAIL and is never masked.
- Corrected misleading comments that claimed the output-token cap fixed the input token-limit error; documented the two guards as independent halves.

Added

- **scripts/tests/test_poe_proxy.sh** — hermetic tests for the Poe proxy: parameters injection, the tool-count cap, built-in-tool preservation, the `P0E_MAX_TOOLS` override, and end-to-end `fix_request`.
- **scripts/tests/test_providers.sh** — hermetic regression tests: the emitted `cma_run_provider` exports the auto-compact window from `CMA_PROVIDER_CONTEXT_LIMIT` only when non-empty; the migration regenerates an outdated wrapper lacking the guard while preserving surrounding alias lines.
- **scripts/tests/proof/claude-live-superpowers-cli.txt** — evidence matrix: every provider alias launched through REAL Claude Code running `/using-superpowers`, classified PASS / FUNDS / BADKEY / NOKEY / FAIL.

v1.11.1 — 2026-07-03 — Live per-alias Claude Code verification (CLI + TUI) + provider fixes

Investigated the reported "most provider aliases fail with an API error." Root cause was NOT a broad implementation defect: launched every alias through real Claude Code (scrubbed env) and probed each provider API directly. 9 aliases pass end-to-end; 2 had genuine model-config drift (fixed below); the remaining failures are all **account-side** (insufficient funds, invalid/expired keys, missing keys, or a plan with no chat model) — not toolkit bugs.

Added

- **scripts/tests/verify_claude_live.sh** — end-to-end live verification of every provider alias through REAL Claude Code in BOTH modes: **CLI** (-p ... --output-format json, authoritative) and **TUI** (driven under a PTY). Each launch runs in a scrubbed env; TUI runs from a throwaway temp cwd so it can never resume a real conversation (the cross-alias .claude.json sync would otherwise auto-resume one). Outcomes are classified **PASS / FUNDS / BADKEY / NOKEY / FAIL** so account problems are never miscounted as toolkit bugs; on a launch FAIL it probes the provider API directly to recover the true cause (e.g. a ccr hang on an upstream 401/429 → BADKEY/FUNDS, not FAIL).
- **scripts/tests/lib/pty_drive.py** — pexpect PTY driver for the interactive Ink TUI (boot, accept any trust prompt, type prompt, capture transcript, quit).
- **scripts/tests/lib/classify_live.py** — shared transcript classifier.

Fixed

- **huggingface** — strong/fast pinned to non-reasoning coder models (Qwen/Qwen3-Coder-480B-A35B-Instruct / Qwen/Qwen3-Coder-30B-A3B-Instruct). The previous models emitted their answer as hidden reasoning_content, returning empty content at low max_tokens and reading as broken. **Verified green** in CLI + TUI.
- **kilo** — strong/fast pinned to verified free-tier models (nvidia/nemotron-3-super-120b-a12b:free / nvidia/nemotron-3-nano-omni-30b-a3b-reasoning:free). The old x-ai/grok-build-0.1 is a **paid** model this key can't access (401 →

100s stall) and the old fast baidu/cobuddy:free is retired (404). **Verified green** in CLI + TUI.

- **inference** — base URL corrected `https://inference.net/v1` → `https://api.inference.net/v1` (old host 301-redirects). NOTE: this key's plan currently exposes no general chat model, so the alias still errors 400 until a chat-capable plan/key is supplied — documented, not silently "fixed."
- **novita-ai** — defensive swap off the retired fast model `sao10K/L3-8B-stheno-v3.2` (404).
- **claude-providers.sh (present_key_vars)** — skip declared-but-**empty** key vars so an empty key (e.g. `SARVAM_API_KEY=`) no longer spawns a broken alias that only errors at launch.

Verified (live on host)

- **9 aliases PASS** end-to-end in CLI (and TUI smoke): `chutes`, `huggingface`, `kilo`, `kimi-for-coding`, `nvidia`, `opencode`, `poe`, `siliconflow`, `zai-coding-plan`.
- Non-PASS are **account-side, not toolkit bugs**: FUNDS — `deepseek`, `fireworks-ai`, `novita-ai`, `openrouter`, `upstage`, `xiaomi`, `zhipuai`; BADKEY — `github-models`, `tencent-tokenhub`; NOKEY — `sarvam`; `plan-limited` — `inference`.
- Sandbox providers suite **113/113**; `shellcheck` clean; python files compile.

v1.10.8 — 2026-07-01 — noclobber-safe router-provider config write

Fixed

- **Every router-transport provider alias broke under `set -o noclobber`**. `cma_run_provider` runs in the user's interactive shell; when that shell has `noclobber` set, the router-config rewrite `jq ... "$cfg" > "$tmp"` failed with *"cannot overwrite existing file"* (the just-created `mktemp` target already exists), silently dropping the update so `claude-code-router` launched with a stale/empty config → API errors. Switched the write to the force-clobber operator `>|`. (`lib.sh`)
- **Existing installs self-heal**: added a regeneration trigger so `cma_ensure_alias_file` rewrites an outdated `cma_run_provider` that lacks the `>|` fix (plain function-body changes previously did not re-trigger on install).

Added (tests)

- `test_providers.sh`: regression asserting the emitted `cma_run_provider` uses `>|` (not a bare `>`) for the router-config write, plus a functional proof that `>` is blocked by `noclobber` while `>|` succeeds.

Verified

- Suite **18/18 green**; shellcheck 0. Root cause reproduced (set -C; echo > "\$(mktemp)" → "cannot overwrite"), fix deployed + confirmed on the host (deployed alias write line = >| "\$tmp").

Note (not a toolkit bug)

- A 46-alias provider sweep shows native providers reach their APIs and return **account-side** errors (e.g. 402 Insufficient Balance) — provider billing/key state, not a toolkit fault.

v1.10.7 — 2026-06-29 — Shared-items drift guard + audit closure

Added (tests)

- **test_unify.sh** — a drift guard asserting claude-unify.sh's SHARED_ITEMS equals lib.sh's CMA_SHARED_ITEMS (used by claude-add-account) minus the intentional CLAUDE.md special-case (unify promotes it via sync_claude_md). Catches the documented hazard of adding a shared item to one list only.
- Cleaned a pre-existing SC2319 lint in test_unify.sh.

Audited — no code changes (3 parallel investigators; every finding independently checked)

- **Unify + rollback engine: clean** — enabledPlugins union, history.jsonl dedup, settings.json merge (malformed-input safe), plugin-manifest path rewrite, and rollback all verified correct across 170+ assertions + edge cases.
- **OpenCode sync (opencode_sync.py): clean** — idempotent + additive verified; two reported "bugs" were refuted against the code: the setdefault "can't update existing keys" IS the documented never-clobber design, and the --enable-all secret nit is not a security issue (an unresolved secret can't leak) and matches "enable everything" semantics.
- **Runtime sync + add-account: clean** — the SHARED_ITEMS "drift" is the intentional CLAUDE.md special-case (now locked by the guard above); sync-state private-key isolation, corrupt-input handling, and add-account idempotency verified.

Verified

- Suite **18/18 green**; shellcheck 0.

v1.10.6 — 2026-06-29 — Committed credential-leak regression test

Added (tests)

- **test_lib.sh** — a committed security regression for `cma_merge_claude_json`: two accounts with distinct `userID/``oauthAccount` and disjoint projects are merged; asserts each account keeps its OWN private auth keys (no cross-account leak in either direction) and the projects subtree is unioned both ways. The function previously had only indirect coverage (via the full unify workflow); this locks the property an audit verified by hand this session.

Verified

- Suite **18/18 green**; shellcheck 0.

v1.10.5 — 2026-06-29 — Provider 'null' field normalization + coverage

Fixed

- **A missing JSON field could write `CMA_PROVIDER_MODEL='null'` (and `TRANSPORT`, alias name) into a provider env file**, launching the provider with a bogus model.
`cma_provider_write_env` normalized `base/fast/context/max` "null"→empty but missed `model` and `transport`; `claude-providers` `multi-sync` also extracted `strong_model/transport/alias_name` with bare `jq -r` (no `// empty`, unlike the already-correct `context_limit/max_output`). Fixed at both the source (`// empty` on every extraction) and the choke point (`normalize model+transport` in `cma_provider_write_env`). Reproduced (`= 'null'`), confirmed fixed (`= ''`).

Added (tests)

- **test_providers.sh** — regression asserting a `null` `model/transport/base/etc.` is normalized to empty; no field ever contains the literal `'null'`.
- **test_session.sh** — EXECUTION tests for the `hint` subcommand (run on every bare launch, previously only string-matched) and for `cma_project_root`'s `git-toplevel` + `symlink` (`pwd -P`) branches.

Audited — no change needed (independently verified, not taken on trust)

- `cma_merge_claude_json`: NO cross-account credential leak; projects unioned; corrupt input skipped gracefully (verified with crafted 2-account inputs).
- BSD/macOS portability: no unguarded GNU-isms (the 3 `readlink -f` hits are comments; `stat -f/-c branch + cma_realpath` guards present).
- `jq` robustness: the `@tsv` sync paths render null as empty (safe); two reported 2>&1 "error-leak" findings were FALSE — there is no 2>&1 on those lines.

Verified

- Suite **18/18 green**; shellcheck 0.

v1.10.4 — 2026-06-29 — set -e/pipefail abort fixes + hardened test coverage

Fixed

- **claude-providers list / remove aborted on a provider with no alias line.** Under `set -euo pipefail`, the alias-name probe `grep ... | sed | head -1` returns 1 (no match) when a provider's `.env` exists but its alias line is absent (manual edit / partial setup); `pipefail` propagated the failure and `set -e` killed the subshell (`list`) or the function before `rm -f` (remove). Guarded both with `|| alias=""`. (`claude-providers.sh`)
- **cma_ensure_alias_file aborted on an alias file lacking export CLAUDE_BIN=.** The `CLAUDE_BIN`-migration probe `grep -m1 '^export CLAUDE_BIN=' ...` returned 1 on an older/hand-edited alias file and aborted the function mid-run under `set -e`. Guarded with `|| _cur_cb=""`. (`lib.sh`)

Changed (tests)

- **test_providers.sh** — replaced the AT-RISK fixed-window `grep -A40 '^cma_run()'` assertions (the push marker had drifted to within 9 lines of the window edge, the same brittleness that already broke -A30 once) with full-body `awk` extraction; added EXECUTION regressions that run the real `claude-providers list/remove` against an alias-less provider and assert no abort.
- **test_coverage.sh** — added a regression that EXECUTES `cma_ensure_alias_file` against an alias file with no `export CLAUDE_BIN=` line and asserts it completes.
- **test_session.sh** — added EXECUTION tests for the `hint` subcommand (run on every bare launch, previously only string-

matched): exits 0, writes only to stderr, names the snake_case project, handles an empty label.

Verified

- Suite **18/18 green**; shellcheck 0. All three aborts reproduced (RED) and confirmed fixed (GREEN); the providers fix proven RED on a guard-stripped copy. Found via 3 parallel investigator subagents, each finding independently reproduced before fixing.

v1.10.3 — 2026-06-29 — Execution-level wrapper test coverage

Added

- **test_wrapper_exec.sh** — the first hermetic test that actually *executes* the generated cma_run wrapper (every other suite only string-matches its emitted body, so a runtime bug — a set -e abort, a dropped unset, wrong call order — could ship past a green suite). It drives cma_run with a stub CLAUDE_BIN env-recorder plus stub claude-session/claude-sync-state, then asserts RUNTIME guarantees: provider-env isolation (a leaked ANTHROPIC_BASE_URL/AUTH_TOKEN/MODEL is genuinely cleared *before* claude runs), session flags reach claude on a bare launch, sync-state pull fires before launch and push after, explicit args pass through verbatim with no session-flag injection, plus a non-vacuity guard proving the stub claude really executed. Proven **RED** on a dropped unset, **GREEN** on the real wrapper.

Verified

- Suite **18/18 green**; shellcheck 0.

v1.10.2 — 2026-06-29 — Self-healing rc source lines + strict rc tests

Fixed

- **Dangling source ".../aliases.sh" lines in rc files.** A transient or moved alias-file path could leave a source line in ~/.bashrc/ ~/.zshrc pointing at a deleted file, so every new login shell printed -bash: .../aliases.sh: No such file or directory. cma_ensure_alias_file now **prunes** any rc source/. line whose aliases.sh target no longer exists (self-heal on the next install), and recognizes an existing source line across ./source and

`$HOME/~`/absolute forms, so re-installs never accumulate duplicate source lines.

Added

- **test_rc_sourcing.sh** (10 strict assertions) — reproduces the bug class the hermetic suite missed (it sandboxes `$HOME` and never inspected or *sourced* the rc files): prune drops dangling / keeps valid + comments + unrelated lines, ensure self-heals, **a fresh shell sources the rc with NO error** (the reported symptom), idempotent (exactly one source line after 3 calls), and cross-form dedup. Proven RED on the old behavior, GREEN on the fix.

Verified

- Suite **17/17 green**; shellcheck 0.

v1.10.1 — 2026-06-29 — Robust cma_run wrapper assertions

Fixed

- **test_claude.sh** used a fixed **grep -A30 window** to scan the `cma_run` body and silently missed the sync-state push marker once the body grew with the v1.10.0 apply-color calls (push slipped past line 30) — failing the suite against a v1.10.0-installed alias file even though the wrapper itself was correct. It now extracts the full function body (awk header → closing brace), robust to future growth.

Verified

- Suite **16/16 green** against the v1.10.0 wrapper; shellcheck 0.

v1.10.0 — 2026-06-29 — Auto-applied per-alias session color + coverage/wiring

The per-alias session color is now **auto-applied** (it was only a hint in v1.9.x), plus self-healing for a stale `CLAUDE_BIN` and several closed test-coverage gaps.

Added

- **Auto-applied per-alias session color.** Each bare alias launch now writes the alias's color into the session as an agent-color record — the exact record Claude Code's `/color` writes — via the new `claude-session apply-color`, called by `cma_run/cma_run_provider` (before launch to colour a resumed session; after exit to colour a freshly-created one). Deterministic `md5(label) mod 8` over red/blue/green/yellow/purple/orange/pink/cyan: each alias gets a stable, distinct colour, and switching the same session between aliases re-colours it. Verified **LIVE** on claude 2.1.195 — written, idempotent, persists across `--resume`. (Prompt-bar rendering must be confirmed visually: `/color` is TUI-only and `claude -p '/color x'` is a no-op, so record injection is the only non-interactive mechanism. See [docs/SESSION_COLOR.md](#).)
- Test coverage: `test_install.sh` (executes `install.sh` in a sandbox — symlinks, alias file, idempotency), `test_verify_scripts.sh` (`model_verify.py`
 - `providers-verify.sh`), `test_session apply-color` tests (incl. the `set -e` regression), `test_coverage B7` (`CLAUDE_BIN` resolver), `B8` (`CLAUDE_BIN` migration), `B9` (`apply-color` wired into both wrappers). `run-proof.sh` now also runs the previously-orphaned `verify_aliases_live.sh`.

Fixed

- **Stale CLAUDE_BIN self-heals.** Existing installs whose alias file pointed `CLAUDE_BIN` at a non-existent path (e.g. `~/.local/bin/claude` where `npm` put `claude` in `~/.npm-global/bin` — the `amber.local` case) now rewrite it to a resolved, executable `claude` on the next `install/ensure`.
- A `set -e/pipefail` bug where `apply-color` aborted before writing on a session that had no existing agent-color record.

Verified

- Suite **16/16 ALL GREEN; shellcheck 0**. Color injection proven **LIVE** on real claude 2.1.195 (write / idempotent / persist-across--resume / recolour-on-alias-switch).

v1.9.2 — 2026-06-29 — Hermetic CLAUDE_BIN resolver test

Fixed

- `test_coverage.sh B7 "fallback when nowhere"` was not **hermetic** and failed on hosts that have a real `/usr/local/bin/`

claude (caught live on thinker.local): the resolver *correctly* returns the system claude there, but the test wrongly assumed "claude nowhere" was achievable under a sandboxed HOME/PATH (it can't mask absolute system paths). Dropped that one assertion; the load-bearing discovery cases (explicit CLAUDE_BIN, ~/.npm-global/bin discovery) stay covered. Runtime behavior unchanged.

Verified

- Suite **14/14 green on all five hosts** (this host, mistborn, thinker, amber, nezha); shellcheck 0.

v1.9.1 — 2026-06-29 — CLAUDE_BIN resolves across per-host install locations

A patch found during the live multi-host rollout of v1.9.0.

Fixed

- **Alias launches failed where claude was installed outside ~/.local/bin.** npm i -g @anthropic-ai/claude-code lands in different prefixes per host (~/.npm-global/bin, Homebrew, ~/.local/bin); the toolkit's fixed CLAUDE_BIN default mis-pointed on those hosts, so every claudeN/provider launch failed "No such file or directory" (amber.local needed a manual symlink to work). cma_resolve_claude_bin now prefers an explicit CLAUDE_BIN, then \$PATH, then the known locations (~/.local/bin, ~/.npm-global/bin, /opt/homebrew/bin, /usr/local/bin), with a ~/.local/bin fallback.

Verified

- Suite **14/14 green; shellcheck 0.** test_coverage.sh B7 covers explicit / npm-global / fallback resolution. v1.9.0's auto-session naming confirmed installed + **live-validated** (create-named + legacy-rename) on all five hosts: this host, mistborn, thinker, amber, nezha.

v1.9.0 — 2026-06-29 — Per-project auto-sessions that actually work live + zero-coverage tests

A minor release that makes the v1.8.0 "auto session-per-project" feature do what it promised. As shipped, opening any alias gave an **unnamed** session; three root causes — all reproduced and fixed against the real claude 2.1.195 binary, then proven **LIVE end-to-end** — are corrected here, plus test coverage for two zero-coverage utilities and a documentation refresh.

Fixed

- **Per-project auto-session naming never actually named the session — now proven LIVE.** Three independent root causes:
 - **Legacy/unnamed sessions were never renamed.** The launcher only `--resume`'d an existing session and never passed `--name`, so a session created by an older wrapper or by plain claude stayed unnamed forever. Fix: always pass `--name` on resume too. Proven live — `claude --resume <id> --name <x>` renames a previously-unnamed session (`custom-title <NONE> → <x>`), contradicting the docs but confirmed empirically.
 - **The session-existence check used a run-collapsing slug.** It collapsed runs of non-alnum to one `-` (`s/[^A-Za-z0-9]+/-/g`), but claude slugs **per char**, so paths with consecutive non-alnum segments (hidden dirs, `/tmp/.private`, `__pycache__`) false-negated the lookup and **re-created** instead of resuming. Fix: per-char slug (`s/[^A-Za-z0-9]/-/g`), matching the real on-disk dir names.
 - **cma_run self-heal regenerated only on a missing unset ANTHROPIC_ marker.** Wrappers predating auto-session carried that marker but not the claude-session one, so they never regained the integration. Fix: regenerate when **either** marker is missing.

Added

- **docs/SESSION_COLOR.md** — resolves the previously dangling reference, documenting per-project auto-session naming and the honest `/color` limitation: in claude 2.1.195, `/color` is **TUI-only** (no CLI flag, no `settings.json` key, no env var — verified against the binary and the docs), so the toolkit can only print a deterministic per-alias hint, never auto-apply it.
- **test_toon.sh (9 assertions) and test_bootstrap.sh (39 assertions)** — both utilities previously had **zero** coverage. `toon` (hermetic, SKIP-if-no-node): `toon.mjs` encode/decode round-trip, the `toon_encode.py` `python→mjs` chain, and non-zero

exit on invalid JSON. bootstrap (hermetic): `claude-bootstrap --count 2 --yes` in a sandbox `$HOME` asserting account dirs, shared symlinks, private-file isolation, alias lines, and the documented refuse-to-clobber re-run behavior.

- **test_coverage.sh B6** asserts the emitted `cma_run / cma_run_provider` bodies actually carry the auto-session integration (bare-launch guard, `claude-session` flags, `eval set -- apply, color hint`) — the session script's own unit tests can't see the wrapper — plus a **self-heal regression**: a stale `cma_run` missing the `claude-session` marker is regenerated (exactly one `cma_run()`, provider-env isolation retained, aliases preserved). `test_session.sh` updated for `--name-on-resume` and a per-char-slug regression (a `/.cfg/` path must resume, not re-create).
- **Docs refresh.** README rewritten as a project landing page; `scripts/README` refreshed with the full current script inventory; the long-form guide gained "Per-project auto-session & per-alias color" and "TOON utility" sections; the `/color` notes pinned to the verified `claude 2.1.195` (superseding older `2.1.178` references).

Verified

- Full suite **14/14 ALL GREEN**; **shellcheck 0**. The session fix proven **LIVE end-to-end** on the real `claude 2.1.195` binary — fresh create, resume, and legacy-unnamed rename all confirmed. New tests are non-vacuous (concrete expected values / negative controls). Installed live on this host and validated.

v1.8.1 — 2026-06-29 — Merge-engine correctness + portability hardening

A patch release: an adversarial correctness audit of `claude-unify`'s merge engine plus a BSD/GNU portability pass over the test + proof tooling. All fixes/hardening — **no new features**.

Housekeeping: a divergent mirror lineage that re-created `v1.7.11` (`1e975e5`) was merged back into main resolved to **OURS** (local already carries `v1.7.11` → `v1.8.0` and later fixes that supersede it), leaving a tree byte-identical to HEAD so all four mirrors converge on one lineage; the `containers` submodule was fast-forwarded to latest main (`71d3256` → `67ed35a`).

Fixed

- **history.jsonl merge fused records across a source missing its trailing newline.** `merge_history_jsonl` cat'd sources into a temp first, gluing one file's last line onto the next file's first line → two entries collapsed into one invalid-JSON line. Fix:

feed files straight to `awk` (fresh record per file). Regression **R1** (RED before, GREEN after).

- **enabledPlugins union dropped "any true".** The `jq` used `+/*` (rightmost-wins), so a plugin enabled in an earlier account but false in the lexically-last account ended up disabled for everyone — contradicting the documented "any true survives" guarantee. Fix: `OR-of-true` reduce over every account. Regression **R2**.
- **A single malformed settings.json aborted the whole unify — and naive guarding then risked silent config loss.** The multi-file `jq -s` ran unguarded under `set -e` (settings is item 15 of 16), halting mid-run. Merely skipping the merge was worse: `link_to_shared` still replaced each valid account's real settings.json with a symlink to a never-written target (a dangling link → silent loss, exit 0). Final fix (hardened after adversarial review): validate each file with `jq empty` and merge only the valid ones (a malformed sibling is excluded, not fatal), and `link_to_shared` refuses to create a link when the shared target is absent. Regression **R3** (asserts the valid account's settings stay readable, not just that unify exits 0).
- **Directory-merge conflicts were resolved by lexical account name, not recency.** `merge_dir_into_shared`'s second pass overlaid only `ACCOUNTS[-1]` (alphabetically-last) while claiming to bias toward the "most recently active" account, so a stale account sorting last could clobber fresher memory/*.`md`. Fix: overlay every account with `rsync -au` so the newest-mtime file wins each conflict, independent of name/order. Regression **R4**.
- **Rollback left dangling symlinks.** Unify symlinks every shared item into each account; for an item an account never had there is no `.preunify` backup, so rollback's restore loop never visited it and the symlink dangled once `SHARED_DIR` moved aside. Fix: after restoring backups, remove any leftover symlink whose target points into `SHARED_DIR` (skipping the shared store itself). Regression **R5**.
- **Rollback restored a non-deterministic backup.** `find -print0` was unsorted, so when a path had several `.preunify.*` backups an arbitrary one was restored. Fix: `sort -z` (timestamps are `YYYYMMDDHHMMSS` = lexical-chronological) so the oldest — the true pre-unify original — wins. Regression **R6**.
- **test_unify.sh B2 was a vacuous PASS.** It called `cma_realpath` (a `lib.sh` function) without sourcing `lib.sh`, so the call errored to empty and the assertion compared `"" == ""` — the symlink target was never verified. Fix: `source lib.sh + set +e` (matching every sibling test that uses `lib` functions directly). Now prints the real resolved `SHARED_DIR/plugins/cache` path.
- **Portability: 3 GNU-only constructs broke the test/proof tooling on macOS** (the shipped runtime toolkit was already clean). `readlink -f` (no `-f` on BSD) in `assert_symlink_to/` `test_unify.sh` returned empty → spurious symlink pass/fail, fixed with a self-contained `_assert_realpath` in `assert.sh` + `cma_realpath` in `test_unify.sh`; `sed -E 's/\x1b...//'` (`\xNN` is GNU-

sed-only) in run-proof.sh/verify_opencode_live.sh left ANSI in, skewing grep -c counts, fixed by building the ESC byte via printf '\033'; unguarded timeout (GNU coreutils) in verify_opencode_live.sh, fixed by resolving timeout/gtimeout once and degrading if absent.

- **De-vendored node_modules.** node_modules/@toon-format/toon was committed by an accidental "Auto-commit" yet load-bearing (scripts/toon.mjs imports the bare specifier; toon_encode.py shells out to it) with nothing ever running npm install. Removed from the tree (git rm --cached + gitignore / node_modules/). Proven by fresh-clone simulation: ERR_MODULE_NOT_FOUND before, encodes correctly after.
- **mktemp portability.** Standardized every bare mktemp [-d] to the templated mktemp [-d] "\${TMPDIR:-/tmp}/cma.XXXXXX" form CLAUDE.md prescribes (BSD mktemp requires a template; only GNU tolerates a bare call) across lib.sh, claude-unify/providers/opencode-sync/session/install, and the test harness (the sandbox keeps its cma-test. prefix for the cleanup safety check).

Added

- **B5 token-limit guard coverage** (test_coverage.sh). The v1.8.0 context_limit/max_output path (cma_provider_write_env → CMA_PROVIDER_CONTEXT_LIMIT/CMA_PROVIDER_MAX_OUTPUT → cma_run_provider exporting CLAUDE_CODE_MAX_OUTPUT_TOKENS) shipped with **zero** tests — the only v1.8.0 fix lacking one. 4 cases / 6 concrete-value assertions: round-trip (262144/32768), null→empty normalization, 7-arg back-compat, and the emitted wrapper carrying the export.
- **npm install step in install.sh** (soft — warns, never hard-fails, when npm is absent; core unify/add-account needs no Node), so a fresh clone gets @toon-format/toon without a vendored tree. curl-install.sh inherits it via delegation.
- **+16 regression assertions** — 6 in test_coverage.sh (B5) and 10 in test_unify.sh (R1-R6 above), each written RED-before / GREEN-after.
- **Documented two deliberate merge/sync trade-offs** in-code so they are explicit rather than silent: cma_merge_claude_json replaces (not element-unions) array values; claude-sync-state pull/push is last-writer-wins (no portable mutex is worth its stale-lock failure modes for a per-launch hook).

Verified

- Full suite **12/12 ALL GREEN; shellcheck 0**; all .py compile under python3 -W error. Each bugfix proven **RED-before / GREEN-after**; the de-vendor proven via fresh-clone simulation (node scripts/toon.mjs + toon_encode.py); ESC-strip verified functionally; the post-merge tree confirmed byte-identical to

HEAD. Installed live on this host and validated against all existing aliases (3 native + 44 provider) + `claude-list-accounts`.

v1.8.0 — 2026-06-29 — Alias isolation + token-limit guard + per-project auto-sessions

A systematic-debugging pass fixing three reported issues plus a new session-per-project feature. Every root cause was reproduced and the fix proven with physical evidence before shipping.

Fixed

- **CRITICAL — aliases cross-contaminated API endpoints across sessions.** `cma_run_provider` exports `ANTHROPIC_BASE_URL/` `AUTH_TOKEN/MODEL/` `SMALL_FAST_MODEL` into the interactive shell, and native `cma_run` did **not** clear them — so running a provider alias (e.g. `xiaomi`) and then a native alias (`claude1`) in the same shell made the native one inherit the provider's endpoint (`api.xiaomimimo.com`). `cma_run` now unsets those four vars before launch. Proven live: after a leaked `xiaomi` env, native launch shows `ANTHROPIC_BASE_URL=<unset>`. Existing installs auto-regenerate the wrapper (migration keyed on the new unset `ANTHROPIC_marker`).
- **Token-limit 400 ("exceeded model token limit: 262144").** The `models.dev` catalog's per-model `limit.context /` `limit.output` were read for ranking but never emitted, so Claude Code overshot a provider's real context window. `providers_resolve.py` now emits `context_limit + max_output`; `providers_generate.py` and `cma_provider_write_env` write `CMA_PROVIDER_CONTEXT_LIMIT / CMA_PROVIDER_MAX_OUTPUT` into each `.env`; and `cma_run_provider` exports `CLAUDE_CODE_MAX_OUTPUT_TOKENS` from it. Proven: `kimi-for-coding` now resolves `context_limit=262144` `max_output=32768` from the live catalog.
- **"workspace has not been trusted" warning on launch.** Confirmed NOT a merge bug (trust propagates across accounts correctly); the warned project was simply never trusted under any account. The launch wrapper now writes `projects[<root>].hasTrustDialogAccepted=true` for the launching project via the new `claude-session` helper.

Added

- **Auto session-per-project (`claude-session.sh`).** Every bare alias launch (native or provider) now resumes — or, the first time, creates — one long-lived Claude session per project root:

stable --session-id (UUID derived from the git-root path), --name set to the root dir basename in lowercase snake_case (Android 15 → android_15). Explicit args/flags are always respected verbatim. Verified against the real claude CLI: --session-id creates, --resume resumes.

- **Per-alias color hint.** A deterministic alias→color mapping over Claude Code's real palette (red blue green yellow purple orange pink cyan); printed as a /color <x> tip on launch. (Investigated thoroughly: /color is a TUI-only command with no CLI flag / settings key / writable persistence, so it cannot be auto-applied — the toolkit suggests it rather than faking it.)
- **test_session.sh** — 27 hermetic assertions for name/id/color/flags/trust/ git-root behavior. **run-all.sh is now 12 files / 60 assertions, ALL GREEN.**

Verified

- Full suite **12/12 ALL GREEN; shellcheck 0**; all .py compile under python3 -W error. All four items proven end-to-end against the live catalog and the emitted alias file.

v1.7.12 — 2026-06-28 — One-line curl installer

Added

- **curl-install.sh** — one-line bootstrap installer:

```
curl -fsSL https://raw.githubusercontent.com/vasic-digital/claude-toolkit/main/scripts/curl-install.sh | bash
```

Detects platform (Linux/macOS) and shell, auto-installs missing hard dependencies (jq, rsync, awk) via the system package manager (apt/dnf/apk/pacman/brew), clones (or pulls if already present) the repo with all submodules recursively to ~/claude-toolkit, runs install.sh, and prints next-steps.

Idempotent; re-runnable. Install dir overridable via CLAUDE_TOOLKIT_DIR env var.

- **README.md** — curl one-liner added at the top of the Install section.
- **test_curl_install.sh** — 22 hermetic tests covering syntax, permissions, URL correctness, submodule cloning, idempotency, platform detection, dependency checks, error handling, and next-steps output.

Verified

- `bash -n + shellcheck 0` on `curl-install.sh` and `test_curl_install.sh`.
- `run-all.sh` **11/11 ALL GREEN** (was 10; `+test_curl_install.sh`).

v1.7.11 — 2026-06-28 — Round-4: coverage-gap regression tests, toon recursion guard, arg validation

Fourth audit round: found the codebase is converging (export-docs, test harness, add/list/remove/rollback all verified clean); shipped 10 targeted coverage tests closing the same shallow-coverage class that let the v1.7.10 enable-plugins bug ship; plus 2 LOW code findings.

Fixed

- **toon_encode.py fallback_encode**: unbounded recursion on deeply nested JSON. A crafted input with >1000 levels would blow Python's default stack and exit with an unhandled traceback instead of encoding. Added a `_depth` guard; beyond 64 levels emits compact JSON as a safe fallback.
- **toon.mjs encode-file / decode-file**: running `toon.mjs encode-file` with no argument produced a confusing `TypeError` from Node's `fs.readFileSync`. Now prints `Error: encode-file requires a filename argument + exit 1`.

Added — coverage-gap regression tests (the same class that let enable-plugins

ship)

- **B9 (HIGH) — cma_ensure_alias_file migration path** (`test_coverage.sh`): builds a realistic old `cma_run_provider()` body lacking `claude-sync-state`, calls `cma_ensure_alias_file`, asserts the body is migrated, the following alias `claude1=` survives, and `cma_run_provider()` appears exactly once.
- **B3 (HIGH) — _cma_q bash quoting in cma_provider_write_env** (`test_coverage.sh`): sources a `.env` with a model name containing a literal single quote and asserts it round-trips intact; also asserts an injection payload does NOT execute on source (mirrors the already-tested Python `q()`).
- **B1 (HIGH) — absorb_default_plugins** (`test_unify.sh`): creates a real plugin file under `$HOME/.claude/plugins/cache/` before unify; asserts it lands in `$SHARED_DIR/plugins/cache/`.

- **B2 (HIGH) — link_default_plugin_subdirs** (test_unify.sh): asserts \$DEFAULT_DIR/plugins/cache becomes a symlink into \$SHARED_DIR/plugins/cache after unify, and that re-running unify doesn't create a second backup.
- **B4 (MEDIUM-HIGH) — sync_claude_md seed branches** (test_unify.sh): branch (b) seeds \$DEFAULT_DIR/CLAUDE.md and asserts it wins; branch (c) removes it and gives an account a CLAUDE.md, asserts that one wins.

Verified

- run-all.sh **10/10 ALL GREEN** (coverage now 39+10=49 assertions; unify now 43+7=50); **shellcheck 0**; all .py compile under python3 -W error; node --check toon.mjs clean; toon_encode 500-level-nest no longer crashes; toon.mjs missing-arg gives clean error + exit 1.

v1.7.10 — 2026-06-28 — Round-3 audit: enable-plugins bug fix, path-traversal guards, proxy robustness

Third audit round (deep dive on the less-covered surface: opencode_sync, claude-unify merge, the poe proxy, bootstrap). Fixes verified centrally.

Fixed

- **cma_enable_plugins silently enabled NO plugins when given 3 or more** (lib.sh). The jq --arg index was derived as \$ {#args[@]}/2, but each iteration appends **three** elements — so for the default 4 always-on plugins it produced arg names p0,p1,p3,p4 while the jq program referenced \$p0..\$p3; \$p2 was undefined, jq failed, 2>/dev/null swallowed the error, and enabledPlugins was left empty. Replaced the derived index with a dedicated counter. Proven live: cma_enable_plugins a b c d now yields all four true (was empty); a ≥3-plugin regression test was added.

Security (defense-in-depth)

- **Path traversal via unvalidated provider id** in claude-providers.sh cmd_show / cmd_remove: \$id was interpolated into <dir>/\$id.env and then cat/rm -f'd without validation. Now rejected unless it matches [A-Za-z0-9._-] (blocks ../), matching cma_provider_write_alias.
- **opencode_sync.py \${CLAUDE_PLUGIN_ROOT} path traversal**: a malicious installed plugin could set an arg like \$

{CLAUDE_PLUGIN_ROOT}/../../../../tmp/evil.js and --enable-all-local would have OpenCode exec the traversed path.
Expansion now lexically contains the result to the plugin dir; an escaping value is left unexpanded (fails safe).

- **cma_write_alias now rejects whitespace in the config dir** (lib.sh): an unquoted space silently word-split the alias into a bogus command — now a clear error instead of a broken alias.

Robustness

- **poe_proxy.py**: resolve_refs gained a recursion-depth guard (a circular \$ref previously crashed the request handler with RecursionError); the success-path gzip.decompress is now guarded like the error path, so a corrupt gzip body no longer propagates.

Verified

- run-all.sh **10/10 ALL GREEN** (coverage now 32 assertions incl. the enable-plugins + injection regressions); **shellcheck 0**; all .py compile under python3 -W error.
- cma_enable_plugins fix proven live (4 plugins → all true); opencode containment + id validation proven with PoCs. The model-verification / alias- write path is unchanged from v1.7.9's live-proven 137 models / 32 aliases.

Audit (round 3) — verified clean

cma_merge_claude_json private-key isolation, eval-token provenance, cma_validate_alias, proxy bind (localhost only) + no key logging, _cma_q escaping, merge_settings_json atomic write, history dedup, rollback NUL-safe traversal, bootstrap --dir-of injection filter. (opencode_sync --enable-all intentionally bypasses the needs-secret guard — operator opt-in, documented.)

v1.7.9 — 2026-06-28 — Hardening round 2: injection-safe alias writes, broadened secret redaction, docs accuracy, shellcheck 0

A second multi-agent audit + hardening pass on top of v1.7.8 (adversarial security audit + doc-accuracy audit + lint sweep, fixes verified centrally).

Security

- **Provider id / config dir can no longer inject shell via the alias file** (`lib.sh cma_provider_write_alias / cma_write_alias`). Both interpolate values into `alias name="..."` lines that the shell **re-parses on invocation**, and `jq @tsv` does not escape `"`. They now reject shell metacharacters (provider id restricted to `[A-Za-z0-9._-]`; config dir rejects `" $ \ \ ; & | < > ( )` and newline). Proven: `afoo"`; `touch ...`` payload is rejected, no command runs, the hostile alias is never written.
- **Keys-file read no longer breakable by a quote in the path** (`claude-providers.sh cmd_sync_multi`). The old bash `-c "set -a; source '$keysf'; ..."` let a single quote in the keys-file path break out of the string. Replaced with an isolated subshell `( set +e; set -a +u; . "$keysf"; set +a; eval ... )` — the same safe pattern `cmd_sync` already used. Proven with a `do n't/` path.
- **.env value quoting** (`providers_generate.py`): `q()` now POSIX single-quote-escapes embedded quotes (mirrors `lib.sh _cma_q`), so a catalog value containing a quote can't inject when `cma_run_provider` sources the `.env`. Proven: an injection payload is neutralized to a literal.
- **xtrace secret leak** (`lib.sh cma_run_provider`): the indirect key read is now wrapped in `set +x/restore` so an active `set -x` in the user's shell can't echo the key to the terminal or a redirected log.
- **Broadened secret redaction + guard**: `cma_redact_secrets()` (`verify_opencode_live.sh`) and the committed-proof scan guard (`test_lib.sh`) now also catch `sk-ant-`, `hf_`, `AIza`, `xoxb-/xoxp-/xoxs-`, `pc-`, `re_`, `secret_`, and JWTs — regardless of JSON field name — closing the gap where arbitrary MCP env-var names (e.g. `NOTION_API_KEY`) slipped through the original six-name allowlist.

Fixed

- **install.sh used readlink -f** (absent on BSD/macOS) for its symlink up-to-date check — missed by the v1.7.7 sweep. Now uses `cma_realpath`; the `test_lib.sh` guard scans `install.sh` too.
- **verify_aliases_live.sh hardcoded one developer's account dirs**, producing false FAILs on every other host. Now discovers accounts dynamically and skips dirs that don't exist.
- Dead code / cruft: `providers_generate.py` (unused `import`, dead vars, `lambda`→`def`, a no-op `provider_id + ('' if ... else '')`); `model_verify.py` (unused `import hashlib`); `model_verify.py` `docstring --key` → `CMA_PROBE_KEY`.

Docs

- Long-form doc + READMEs + `CLAUDE.md` corrected against the code: macOS rc-file caveat (`~/ .zshrc` only), the test table now

lists all 10 suites, the full installed-command list (+claude-providers/claude-sync-state/ claude-bootstrap), repo-relative paths (was ~/Documents/scripts/), a new claude-bootstrap section, the CMA_PROBE_KEY security model in §11, and a refreshed date stamp.

Quality

- **shellcheck: 93 → 0** across all scripts. Added .shellcheckrc (external-sources=true) which resolves the sourced-file warnings properly; fixed the \$?-after-condition (SC2319) test idioms, SC2015/SC1090/SC1003; the one remaining reserved no-op flag carries a justified inline disable.

Verified

- scripts/tests/run-all.sh **10/10 ALL GREEN**; **shellcheck 0**; every .py compiles under python3 -W error.
- Injection PoCs (provider id, config dir, .env value, keys-file path) all proven neutralized; broadened redaction proven against AIza/hf_/JWT/etc.
- Live sync --multi: **137 models verified, 32 aliases** across 8 providers (opencode 4, poe 33, chutes 7, huggingface 6, nvidia 30, openrouter 14, siliconflow 38, xiaomi 5), zero CMA_PROBE_KEY/unbound errors — identical to the v1.7.8 baseline, so the new key-read path is non-regressive end-to-end.
- 4-host byte-parity + 10/10 suite re-verified after deploy.

v1.7.8 — 2026-06-28 — Secret hygiene (argv + committed-proof leaks), dead-code fix, coverage tests

Security + robustness follow-up found by a parallel multi-agent audit of v1.7.7. Four independent subagents fixed disjoint file sets; integration + the full suite + live multi-model verification were run centrally.

Security

- **API key no longer passed on argv** (model_verify.py + claude-providers.sh). cmd_sync_multi invoked model_verify.py --key "\$token", placing the secret verbatim in /proc/<pid>/cmdline and ps aux output — readable by any user on a multi-user host. The key now flows via the CMA_PROBE_KEY environment variable (set per-command, not exported); model_verify.py reads it from the environment and errors clearly if unset. The --key flag is removed entirely.

- **API key no longer passed to curl on argv** (verify_aliases_live.sh). Six live-probe calls used -H "Authorization: Bearer \$key". The header is now written to a mktemp'd, chmod 600 config file consumed via curl --config (portable on GNU + BSD curl) and removed via an EXIT/INT/TERM trap.
- **Leaked secrets purged from committed proof artifacts** (committed in 24bc379, rolled into this release): the OpenCode live verifier wrote resolved opencode debug config / mcp list output — which contained a real provider key and a DB connection-string password — verbatim into the committed proof dir. The three artifacts are redacted; the generator (verify_opencode_live.sh) now redacts via cma_redact_secrets() before writing (raw dump → .raw temp → redacted file → .raw removed). **Operator follow-up still required:** rotate the leaked key and decide on a git-history scrub — the values remain in history on all four remotes.

Fixed

- **Unreachable code** in verify_aliases_live.sh: exit \$failed sat *before* the Claude-alias test function and its caller, making them dead (shellcheck SC2317). exit \$failed moved to the final statement.
- **Fragile \$? capture** in test_list.sh: grep ...; [[\$? -ne 0]] then assert_eq 0 \$? read \$? from the wrong command. Now captures rc=\$? immediately.
- **Unquoted glob** in claude-sync-state.sh:67: "\$HOME"/\${ACCOUNT_PREFIX}prov-* / → "\$HOME/\${ACCOUNT_PREFIX}"prov-* / so only the intended * globs.
- **SyntaxWarning: invalid escape sequence '\ ' in** providers_resolve.py: the usage docstring's \ line-continuations are now a raw string (r""").

Added

- **test_coverage.sh** — 11 new hermetic tests (19 assertions) covering previously-untested lib.sh behavior: cma_ensure_alias_file (fresh / idempotent / old-format migration preserving unrelated lines), cma_can_prompt (CMA_NONINTERACTIVE=1 and no-tty both non-interactive), cma_enable_plugins (JSON shape + additive + the jq // falsy-vs-null upgrade), cma_link_shared_items (every CMA_SHARED_ITEMS entry becomes a symlink into \$SHARED_DIR, idempotent), and stats-cache.json newest-by-mtime selection.
- **Proof regression guard** (test_lib.sh): scans scripts/tests/ proof for provider-key prefixes and URL user:password@ creds, counting suspect lines so a failure never re-echoes a secret.

Verified

- `scripts/tests/run-all.sh` — **10/10 ALL GREEN** locally (was 9; +`test_coverage.sh`).
- Live multi-model verification (`claude-providers.sh sync --multi`, real HTTP probes with the host's real keys): **137 models verified, 32 aliases generated** across 8 providers (opencode 4, poe 33, chutes 7, huggingface 6, nvidia 30, openrouter 14, siliconflow 38, xiaomi 5). Zero `CMA_PROBE_KEY`-unset and zero unbound variable errors — the `env-var` key path works end-to-end. (Providers with 0 verified are external: dead/paid keys, HTTP 401/402/403, WAF blocks — not toolkit regressions.)
- The new proof secret-scan guard immediately earned its keep: on first cross-host run it flagged a **stale, pre-redaction proof dir on all three remote hosts** (3 files with literal secrets), which were then re-synced with the redacted artifacts.
- `model_verify.py` / `providers_resolve.py` compile clean under `python3 -W error`.

v1.7.7 — 2026-06-28 — Portable realpath (BSD portability hardening), set -u edge fix, regression tests

Follow-up hardening release found by a parallel multi-agent audit of v1.7.6.

Fixed

- `readlink -f` → **portable cma_realpath** at three sites: `claude-unify.sh` (`already_linked_to_shared` and `merge_settings_json`) and `claude-list-accounts.sh` (the link check). `readlink -f` is absent on older macOS and on other BSDs (FreeBSD/NetBSD); there the checks silently fail — making `claude-unify` re-link every shared item on each re-run (accumulating stale `.preunify.*` backups) and `claude-list-accounts` report linked accounts as "not linked". **Honest scope:** modern macOS (Sequoia) and GNU coreutils DO support `readlink -f`, so on the current fleet this was a *latent* bug with no active symptom — but it broke the toolkit's stated BSD portability. Replaced with a new pure-bash `cma_realpath` (single-arg `readlink symlink-walk + pwd -P`), verified to produce output identical to `readlink -f` on macOS.
- **set -u empty-array edge in cma_enable_plugins** — `jq "{args[@]}"` with an empty `args` is an "unbound variable" error on bash 3.2 (reachable via `CMA_ALWAYS_ON_PLUGINS=""` from the non-re-exec'd `claude-providers.sh`). Guarded with `${args[@]}+"${args[@]}"`.

Added

- **cma_realpath** portable canonicalizer in `lib.sh`.
- **Regression tests** (`test_lib.sh`): `cma_realpath` resolves a symlink chain and is identity on a real path; plus a guard asserting NO runtime script *invokes* `readlink -f`.

Verified

- `scripts/tests/run-all.sh` — **9/9 ALL GREEN on all four hosts**: nezha, thinker, amber (Linux), mistborn (macOS, re-exec to bash 5.3, BSD userland).
- `cma_realpath` output confirmed byte-identical to `readlink -f` on macOS.

Audit findings (v1.7.6 — no code change required)

- Disabled providers are EXTERNAL, not toolkit bugs (toolkit correctly disabled them on failed verify): `github-models` → HTTP 401 (dead GitHub PAT), `upstage` → HTTP 403 from AWS WAF (egress-IP block).
- `api_keys.sh` across all 4 hosts: **0 dangling refs, 0 duplicates, 0 malformed**; key parity confirmed (mistborn's 2 host-local Kimi-Platform keys preserved).
- Cross-host integrity: all 11 toolkit scripts byte-identical to the released tag on every host.
- Known/deferred: published tags `v1.2.0` (gitlab) and `v1.5.0` (gitlab/gitverse/gitflic) point to older commits than local — reconciling needs a force tag push; left for a maintainer decision.

v1.7.6 — 2026-06-28 — Always-non-interactive execution, alias-file integrity, macOS/bash-3.2 portability, 4-host rollout

Fixed

- **Alias-file corruption from a mis-firing migration** — `cma_ensure_alias_file`'s "outdated `cma_run_provider`" migration grepped for `claude-sync-state pull`, but the emitted on-disk text is `.../claude-sync-state" pull` (a quote precedes the space), so the guard **never matched** and the migration fired on *every* alias write. Its `awk` then chopped everything from `cma_run_provider()` to EOF — destroying previously-written provider aliases and any `claudeN` aliases that follow the function block. This silently corrupted the alias file on multi-provider /

multi-account hosts. Detection is now scoped to the function body and matches the bare command name (quote/space agnostic), and the migration removes **only** the function block, preserving alias lines. This was the single root cause of the failures across `test_providers.sh`, `test_claude.sh`, and `test_add_remove.sh`.

- **set -u abort while sourcing the keys file** — provider sync sourced `~/api_keys.sh` inside a `set -euo pipefail` subshell. A dangling reference in the user's keys file (e.g. `export SARVAM_API_KEY=$ApiKey_Sarvam_AI_India`) aborted the source **mid-file** under `nounset`, leaving every key defined *after* it unexported — so those providers silently failed verification ("unverified") and stderr was spammed with "unbound variable". Keys are now sourced with `nounset` disabled (subshell-local in sync; save/restore around the alias-file `cma_run_provider`). Installed alias files are auto-migrated to the `nounset-safe` wrapper on next sync.
- **macOS / bash-3.2 portability of the test harness** — `tests/run-all.sh` used `mapfile` (bash 4+), so the **entire suite failed to run on stock macOS**. Replaced with a portable read loop and guarded empty-array expansion under `set -u`. Same fix applied to `test_lib.sh` and `tests/lib/sandbox.sh` (empty `${arr[@]}` expansions are unbound on bash 3.2). The suite now runs green on macOS bash 3.2.

Added

- **CMA_NONINTERACTIVE + automatic TTY detection** — a new `cma_can_prompt` helper makes every prompt (`claude-add-account`, `claude-remove-account`, `claude-bootstrap`) fall back to its non-interactive default whenever no terminal is available (CI, SSH without a PTY, the test sandbox) or when `CMA_NONINTERACTIVE=1` is exported. Toolkit execution is now **always non-interactive off a terminal**. Destructive account removal still refuses (rather than guessing) without `--yes` when it cannot confirm.
- **Regression tests** for non-interactive `claude-add-account` and for alias-line survival across repeated account adds.
- **test_export.sh graceful SKIP** when its prerequisites (pandoc + a PDF engine) are absent — matching the existing SKIP convention for optional-dependency features.

Multi-host rollout (nezha · mistborn.local · thinker.local · amber.local)

- Distributed `~/api_keys.sh` to every host via a **no-loss merge** (host-local keys preserved — e.g. mistborn kept its 2 Kimi-Platform keys; amber created fresh) and wired **both** `.bashrc` and `.zshrc` to source it on every host.

- Installed/updated the toolkit on all four hosts and configured claude1/claude2/claude3 on each; installed Claude Code on amber.
- Ran live provider/model detection on every host — **17-20 active providers each**, models verified via HTTP probes, **0 unbound errors**.

Verified

- scripts/tests/run-all.sh — **9/9 files, ALL GREEN on all four hosts**: nezha (Linux), thinker (Linux), amber (Linux), mistborn (macOS / bash 3.2).
- Cross-host: both rc files source api_keys.sh; claude1/2/3 + poe/deepseek/xiaomi aliases present on every host.

v1.7.5 — 2026-06-28 — Cross-provider / resume session visibility fix

Fixed

- **Cross-provider /resume session loss** — when switching between provider aliases (e.g., deepseek → opencode → kimi-for-coding), /resume would sometimes show empty session history. Root cause: the cma_run_provider function in the alias file was **missing sync-state pull/push calls** that were present in lib.sh. The alias file is what actually runs when a user invokes an alias, so the sync never happened.
- **Migration for outdated alias files** — added automatic detection and regeneration of outdated cma_run_provider functions in lib.sh. If the function exists but lacks claude-sync-state pull, it's removed and rewritten with the correct implementation.
- **Router transport transformer config** — added transformer: {use:["cleancache","streamoptions"]} to the alias file's router transport section (was only in lib.sh), ensuring cache_control stripping works for all router-transport providers.

Root Cause Analysis

The cma_run_provider function in lib.sh (lines 225-333) correctly includes sync-state pull/push calls, but the alias file's copy of the function was outdated and explicitly stated "cross-account claude-sync-state is intentionally NOT run." This meant:

1. Sessions created under provider A had their lastSessionId written only to A's .claude.json
2. When switching to provider B, B's .claude.json still had its own (different) lastSessionId

3. /resume read B's lastSessionId and couldn't find A's session

After fix: all providers/accounts share the same merged lastSessionId via sync-state.

Verified

- **Local host:** all providers show identical lastSessionId after sync (confirmed)
- **mistborn.local:** 76 projects merged across all accounts/providers (confirmed)
- **Migration:** install.sh correctly detects and fixes outdated alias files on both hosts

Tests

- Cross-alias session visibility (Section 5): **ALL PASS**
- Existing test suite: session-related tests pass

v1.7.4 — 2026-06-26 — Kimi provider fix + AWS IaC MCP disabled by default

Fixed

- **Kimi Code provider base URL** in scripts/providers/overrides.json — changed from /coding/v1 to /coding/ so native transport works correctly.
- **AWS IaC MCP timeout** — removed aws-dev-toolkit/awsiac from the default OpenCode MCP allowlist in scripts/claude-opencode-sync.sh. The server consistently timed out on connection and is now configured but disabled by default.

Changed

- Regenerated Claude_Multi_Account_Fine_Tuning.{html,pdf,docx} from current markdown.
- Refreshed proof artifacts in scripts/tests/proof/.

Tests

- Local: **9/9 ALL GREEN**
- Live OpenCode verification: **9 passed, 0 failed**, 27/27 enabled MCPs connected
- Provider alias verification: **5 passed, 0 failed**

v1.6.6 — 2026-06-21 — TOON integration for token-efficient prompts

Added

- **TOON (Token-Oriented Object Notation)** integration — saves ~40% tokens vs JSON for structured data in LLM prompts by declaring fields once in arrays.
- **scripts/toon.mjs** — Node.js TOON utility (encode/decode/demo)
- **scripts/toon_encode.py** — Python wrapper for TOON encoding
- **docs/TOON_Integration.md** — comprehensive guide on using TOON with Claude Code
- **package.json** — @toon-format/toon v2.3.0 dependency

Token Savings

- File listings: ~39% fewer tokens
- Tool definitions: ~40% fewer tokens
- User records: ~42% fewer tokens
- Accuracy: 76.4% (vs JSON's 75.0%)

Note

TOON formats message CONTENT for token savings. API transport remains JSON (providers require it). HTTP/3 and compression require provider-side support.

Tests

- 8/8 ALL GREEN

v1.6.5 — 2026-06-21 — Poe proxy fix (alias file + install)

Fixed

- **Poe proxy not starting from alias** — proxy logic was only in lib.sh, not in the alias file's cma_run_provider function. The alias file is what actually runs when a user invokes an alias. Added proxy detection + auto-start to the alias file.
- **install.sh: SHARE_DIR → SHARED_DIR** — wrong variable name caused unbound variable error on nezha (Linux, set -u).
- **install.sh: auto-copy proxy scripts** to ~/.local/share/.../proxy/ during install.

Verified

- All 3 Poe aliases work: poe , poe2 , poe3
- Deployed to both local host and nezha.local

Tests

- 8/8 ALL GREEN

v1.6.4 — 2026-06-21 — Poe proxy fix for tool compatibility

Fixed

- **Poe tool format error** — Poe requires parameters in every tool function definition. Claude Code sometimes omits it (valid in Anthropic format, invalid for Poe). Added poe_proxy.py that auto-fixes tools before forwarding to Poe API.
- **Proxy auto-start** — cma_run_provider now auto-starts compatibility proxies for providers that need them (detected by scripts/proxy/<provider>_proxy.py).

Verified

- All 3 Poe aliases work through proxy: poe , poe2 , poe3

Tests

- 8/8 ALL GREEN

v1.6.3 — 2026-06-21 — Poe provider (382 models, 3 aliases)

Added

- **Poe provider** — universal AI platform with 382 models from all major providers. OpenAI-compatible API at <https://api.poe.com/v1>. Chat, code, image gen, video gen, TTS, STT, and more.
- **3 aliases:** poe (claude-sonnet-4.6 + gpt-5.4-mini), poe2 (gpt-5.5 + deepseek-v4-pro-e), poe3 (grok-4 + gemini-3.1-pro)
- **key-aliases:** POE_API_KEY + ApiKey_Poe → poe
- **Tool calling verified** on claude-sonnet-4.6, gpt-5.4-mini, deepseek-v4-pro-e, grok-4

- **382 models categorized:** 130 chat/reasoning, 16 code, 40 image gen, 17 video gen, 12 TTS, 1 STT, 166 other
- **Documentation:** full Poe section in `Provider_Aliases_User_Guide.md`

Verified

- API endpoint responds correctly
- Authentication works
- Tool calling confirmed
- All 3 aliases tested through ccr with "Do you see our codebase?" — all YES

v1.6.2 — 2026-06-21 — Chutes provider documentation + model update

Changed

- **Chutes provider models updated** — catalog was stale. Chutes now offers 13 TEE (Trusted Execution Environment) models. Updated `strong=zai-org/GLM-5.2-TEE`, `fast=Qwen/Qwen3.6-27B-TEE`.
- **Chutes documentation** added to `Provider_Aliases_User_Guide.md` with full model table, TEE explanation, pay-per-use note, and setup instructions.

Verified

- Chutes API endpoint responds correctly
- All 13 TEE models accessible (require funded account for actual inference)
- OpenAI-compatible format confirmed at <https://llm.chutes.ai/v1>

v1.6.1 — 2026-06-21 — cache_control fix + E2E tests

Fixed

- **cache_control parameter error** — Claude Code sends `cache_control` (Anthropic-specific) in its API requests. ccr forwarded this to OpenAI-compatible endpoints which reject it with HTTP 422. Fixed by adding ccr's built-in `cleancache` transformer to every provider config, which strips `cache_control` before forwarding to the provider.

Added

- **alias_e2e_test.py** — end-to-end alias verification script that tests each alias by sending requests through ccr and verifying responses work without errors.

Verified working (all aliases tested with "Do you see our codebase?")

- `opcode` (north-mini-code-free): YES
- `opcode2` (big-pickle): YES
- `opcode3` (nemotron-3-ultra-free): YES
- `deepseek` (native transport): YES
- `deepseek2` (router transport): YES
- `xiaomi` (native transport): YES
- `zai-coding-plan` (router transport): YES

v1.6.0 — 2026-06-21 — Multi-alias provider system

Added

- **Multi-alias provider system** — every provider can now have multiple aliases (`provider`, `provider2`, `provider3`...) exposing ALL working models, not just the top 2. Verified via live HTTP probes with anti-bluff detection.
- **model_verify.py** — comprehensive model verification & scoring engine. Tests every model for a provider via HTTP probes, scores on 7 dimensions (existence 25pts, tool_call 20pts, reasoning 15pts, context_window 15pts, streaming 10pts, latency 10pts, free_tier 5pts). Anti-bluff detection prevents false positives (HTTP 200 with error body, empty responses, boilerplate errors). 24h verification cache to avoid re-testing.
- **providers_generate.py** — multi-alias generation from verified models. Pairs models into alias groups of 2 (strong + fast), handles odd count (last model reused for both positions), single model (used for both positions). Generates env files, shell aliases, and overrides.json entries.
- **claude-providers.sh --multi** — new flag for sync that triggers the full verification + multi-alias generation pipeline. Additional flags: `--max-aliases` (default 5), `--min-score` (default 25), `--verify-concurrency` (default 5).
- **Endpoint normalization** — `/anthropic` endpoints auto-converted to `/v1` for OpenAI-compatible probing during verification.

- **Submodules updated** to helix_translate-2.3.1: LLMsVerifier (ModelVerifier, Seed, xiaomi provider), challenges (anti-bluff §11.4, chaos/stress tests), containers (deploy-stack).

Changed

- Probe max_tokens increased from 32 to 128 — reasoning models need more tokens for chain-of-thought + response (was causing false anti-bluff rejections).
- User-Agent header added to HTTP probes (some APIs require it).

Full test suite

- 8/8 test files passed (ALL GREEN), 0 failures.

Usage

```
# Standard sync (2 models per provider, as before)
claude-providers sync

# Multi-alias sync (verify ALL models, create multiple aliases)
claude-providers sync --multi

# With options
claude-providers sync --multi --max-aliases 10 --min-score 20
```

v1.5.1 — 2026-06-20 — Linux stat fix + nezha deployment

Fixed

- **stat -f %m on Linux** — the mtime cache check in claude-providers.sh used stat -f %m || stat -c %Y as an || chain. On Linux, stat -f succeeds (returning filesystem info, not mtime), so both outputs merged into garbage ("File: ...1781634386"), causing File: unbound variable under set -u. Fixed with case "\$ (uname -s)" to pick the correct flag per platform.

Deployment

- **nezha.local** (Linux x86_64) deployed and verified: 19 providers activated, 100/100 provider tests pass, 5/5 live verifier pass, cross-alias sync confirmed. Evidence in scripts/tests/proof/90-nezha-deployment.txt.

Full test suite

- macOS: 8/8 ALL GREEN
- Linux (nezha): 7/8 pass (export fails: pandoc not installed — pre-existing)

v1.5.0 — 2026-06-20 — Cross-alias session visibility

Added

- **Cross-alias session visibility** — sessions created under ANY alias (claudeN, deepseek, opencode, xiaomi, etc.) are now visible from every other alias via /resume. Memory, project settings, and session data are fully shared across all accounts and providers.
- **claude-sync-state.sh extended** — now discovers provider dirs (~/.claude-prov-*) alongside account dirs for its .claude.json merge. Provider sessions participate in the same lightweight jq merge that keeps account sessions in sync.
- **cma_run_provider sync-state hooks** — the provider wrapper now calls claude-sync-state pull before launch and claude-sync-state push after exit, matching the cma_run pattern. Previously provider sessions were intentionally excluded from sync; now they participate fully.
- **Sandbox test coverage:** 10 new assertions proving cross-alias merge (sessions from account→provider, provider→account, account→account all visible after sync). Providers test 90 → 100 assertions.
- **Live verification:** lastSessionId for a real project confirmed identical across all dirs (3 accounts + 1 provider). 61 projects merged in every .claude.json. Evidence in scripts/tests/proof/80-cross-alias-sessions.txt.

Changed

- scripts/claude-sync-state.sh — provider dirs included in merge targets
- scripts/lib.sh — cma_run_provider wrapper updated with sync-state pull/push
- Alias file aliases.sh — updated cma_run_provider function (re-installed)

Full test suite

- 8/8 test files passed (ALL GREEN), 0 failures. Providers test includes 10 new assertions for cross-alias session visibility.

How it works

1. `claude-sync-state pull` merges every account's + provider's `.claude.json` into the launching dir before Claude Code starts (including `lastSessionId`, `allowedTools`, MCP config, etc.).
2. Claude Code launches with the merged state — `/resume` sees all sessions.
3. `claude-sync-state push` merges the post-session `.claude.json` back out after exit, so the next alias to launch picks up the new session.
4. The `sessions/` directory was already shared via symlink — this release ensures `.claude.json` project settings are also merged.

Performance

- Adds ~1-2 seconds overhead per provider launch (jq merge of `.claude.json` across all dirs). Same overhead that `claudeN` aliases already have.

v1.4.0 — 2026-06-20 — OpenCode Zen provider alias

Added

- **opencode provider alias** — [OpenCode Zen](#) curated AI gateway with **21 free models** (all \$0 cost, all support tool calling + reasoning) and 49 paid models. The alias uses **router transport** (ccr) targeting the OpenAI-compatible endpoint `https://opencode.ai/zen/v1/chat/completions`.
- **Model overrides**: `strong` = `big-pickle` (free stealth model, 200K context, reasoning + `tool_call`), `fast` = `deepseek-v4-flash-free` (free, 200K context, reasoning + `tool_call`). Pinning is deliberate — auto-selection would pick `nemotron-3-ultra-free` (1M ctx) as `strong` and `trinity-large-preview-free` (131K, no reasoning) as `fast`, both suboptimal for coding workloads.
- **key-aliases.json mappings**: `ZEN_API_KEY` → `opencode` and `ApiKey_Opencode_Zen` → `opencode` (both key vars present in the user's keys file).
- **overrides.json pin**: `strong_model=big-pickle`, `fast_model=deepseek-v4-flash-free` (no `transport/base_url` override needed — catalog values are correct).
- **Sandbox test coverage**: resolver tests (key-alias mapping for both key vars, router transport from `@ai-sdk/openai-compatible` npm, `zen/v1` `base_url` from catalog, model override beats auto-selection, stale-model-never-selected guards) + sync e2e tests (env file, alias, config-dir + plugins symlink, account-detection exclusion, idempotency, no-secret-leak). Providers test 69 → 90 assertions.

- **Live endpoint verification:** GET /v1/models HTTP 200; POST /v1/chat/completions round trip HTTP 200 with correct text for big-pickle (stealth, cost=\$0, reasoning_content present) and deepseek-v4-flash-free (cost=\$0); additional free models (mimo-v2.5-free, nemotron-3-ultra-free, north-mini-code-free) all HTTP 200 with cost=\$0. Evidence in scripts/tests/proof/70-zen-live.txt (secret-free).
- **Docs:** dedicated opencode section in docs/ Provider_Aliases_User_Guide.md (full free models table, setup, usage, live-verified notes, stealth model explanation).

Changed

- scripts/providers/key-aliases.json and scripts/providers/overrides.json extended with the opencode entries (config-only; no code changes — same dynamic pattern as Xiaomi v1.3.0 / Z.AI v1.2.0 / DeepSeek).

Full test suite

- 8/8 test files passed (ALL GREEN), 0 failures. Providers test includes 21 new assertions for opencode.

Honest notes

- The alias uses router transport (ccr) because Zen's free models use OpenAI-compatible format (/v1/chat/completions), not Anthropic native format. This adds a ccr dependency that native-transport aliases (deepseek, xiaomi) don't have.
- Big Pickle is a stealth model — the actual model served may vary (observed as deepseek-v4-flash). This is by design per OpenCode's documentation.
- The same pre-existing ~/api_keys.sh set -u issue affects the in-process verifier for all providers; authoritative proof is the direct HTTP round trip.
- The 2 pre-existing, environmental opencode-skill-discovery failures in run-proof.sh remain unchanged (unrelated to this work).

v1.3.0 — 2026-06-19 — Xiaomi MiMo provider alias

Added

- **xiaomi provider alias** — Xiaomi MiMo via the **Anthropic-native endpoint** <https://api.xiaomimimo.com/anthropic> (POST /anthropic/v1/messages). Unlike most providers in this toolkit,

MiMo exposes a genuine native Anthropic endpoint that accepts Authorization: Bearer, so the alias uses **native transport** with no claude-code-router (ccr) dependency — the same direct-launch model as deepseek.

- **Model overrides:** strong = mimo-v2.5-pro (flagship, 1M context, reasoning, tool-call), fast = mimo-v2-flash (256K, cheapest tier). Pinning is deliberate — models.dev lists a mimo-v2.5-pro-ultraspeed id the **live API does not serve**, so the override guarantees only live-served ids are used.
- **key-aliases.json mapping:** XIAOMI_MIMO_API_KEY → xiaomi (the user's key-var name does not match the models.dev provider's documented XIAOMI_API_KEY env).
- **overrides.json pin:** native transport, /anthropic base_url, mimo-v2.5-pro / mimo-v2-flash.
- **Sandbox test coverage:** resolver tests (key-alias mapping, override forces native transport, /anthropic base_url beats catalog /v1, model pinning beats the stale ultraspeed entry, stale-id-never-selected guard) + sync e2e tests (env file, alias, config-dir + plugins symlink, account-detection exclusion, idempotency, no-secret-leak). Providers test 60 → 69 assertions.
- **Live endpoint verification:** GET /v1/models HTTP 200 (10 models); native /anthropic/v1/messages round trip HTTP 200 with correct text for both mimo-v2.5-pro and mimo-v2-flash; tool calling proven (finish_reason: tool_calls
◦ reasoning_content); streaming confirmed. Evidence in scripts/tests/proof/60-xiaomi-live.txt (secret-free).
- **Docs:** dedicated xiaomi section in docs/ Provider_Aliases_User_Guide.md (model table, setup, usage, live-verified notes).

Changed

- scripts/providers/key-aliases.json and scripts/providers/overrides.json extended with the xiaomi entries (config-only; no code changes — same dynamic pattern as Z.AI v1.2.0 / DeepSeek).

Full test suite

- 8/8 test files passed (ALL GREEN), 0 failures. Providers test includes 9 new assertions for xiaomi. Live provider verifier 5/5 PASS.

Honest notes

- The only failures in the repo's run-proof.sh are 2 **pre-existing, environmental** opencode-skill-discovery checks, unrelated to Xiaomi (zero opencode files changed by this release; they fail identically when run standalone).

- The in-process LLMsVerifier step reports (unverified) for every provider because `~/api_keys.sh` has a pre-existing unrelated unbound variable on a different provider's key under `set -u`; authoritative proof is the direct native-endpoint round trip (HTTP 200), recorded in the evidence file.

v1.2.0 — 2026-06-19 — Z.AI Coding Plan provider alias

Added

- **zai-coding-plan provider alias** — OpenAI-compatible router transport via `https://api.z.ai/api/coding/paas/v4` (Coding Max-Yearly Plan endpoint).
- **Model overrides**: `strong` = `glm-5.2` (flagship 1M context reasoning model, free on plan), `fast` = `glm-4.7` (204k context, `tool_call`, 0 cost).
- **key-aliases.json mapping**: `ZAI_API_KEY` → `zai-coding-plan` (targets the coding plan API endpoint instead of the general `z.ai paas` endpoint).
- **overrides.json pin**: overrides auto-selected `strong/fast` models for the coding plan.
- **Sandbox test coverage**: resolver tests (env-key matching, coding endpoint, router transport, `glm-5.2/glm-4.7` model selection) + sync e2e tests (env file, alias, model overrides).
- **Live endpoint verification**: HTTP 200 at `/models` (8 models discovered), curl test of `glm-4.7` chat completion confirmed operational.
- **ccr integration**: provider auto-registered in `~/claude-code-router/config.json` as the active default route.

Changed

- `overrides.json` extended with `zai-coding-plan` section for model pinning.

Full test suite

- 8/8 test files passed (ALL GREEN), 0 failures. Provider tests include 5 new assertions for `zai-coding-plan`.

v1.1.0 — 2026-06-16 — Distributed infrastructure + provider verification

Headline: stand up the full LLMsVerifier System on a remote host for heavy testing against **real production LLM services**, plus end-to-end provider aliases proven on two hosts and two transports.

Added

- **containers + challenges submodules** (submodules/) — the distributed-boot orchestrator and its sibling. `helix-deps.yaml` confirms containers has zero own-org submodule deps.
- **Remote host registration** — `config/containers/nezha.env` registers `nezha.local` as a remote boot/test host (SSH key, podman runtime).
- **LLMsVerifier deployment overlays** (`config/containers/llmsverifier/`):
 - `docker-compose.app.yml` — the `llm-verifier` API (cgo image, config mount, `/api/health` healthcheck, loopback, fail-fast secrets).
 - `docker-compose.infra.yml` — observability tier: prometheus + grafana (auto-provisioned datasource + dashboard) + node-exporter. **No DBs** (the app uses SQLite; postgres/redis were unused and removed).
 - `Dockerfile.nezha` / `Dockerfile.mv` — cgo nested-module builds for the server + the model-verification tool.
 - `patches/0001..0005` — upstream LLMsVerifier fixes (see PR #2 below).
- **Deployment guide** `config/containers/llmsverifier/README.md` and the **Provider Aliases User Guide** `docs/Provider_Aliases_User_Guide.md` (HTML/PDF/DOCX exports included).
- **QA evidence** `docs/qa/20260616-infra/` — verification proofs, endpoint coverage, security posture, observability, per-provider sweeps, dual-host end-to-end alias proofs.

Changed

- **Provider session accent color: orange → purple** across spec, guide, and the long-form doc. (Claude Code 2.1.178 cannot persist a default `/color`, so this is the documented default + a manual `/color purple` — a platform limit.)
- `claude-add-account` consolidated onto the shared `cma_link_shared_items` helper (single `CMA_SHARED_ITEMS` source).
- `claude-export-docs` now also emits **DOCX** (HTML/PDF/DOCX).

Fixed (LLMsVerifier — shipped as PR #2, applied to deployed builds)

- **Auth header missing** — verification requests sent no Authorization header → HTTP 401 for every provider. Now Bearer <key>.
- **cohere 405** — switched to the OpenAI-compatible endpoint (`api.cohere.ai/compatibility/v1`). Verifies at score 1.00.
- **gemini / huggingface** — corrected to OpenAI-compatible / router endpoints (huggingface verifies; gemini code-ready pending a valid key).
- **model-id strictness** — verifies a requested id directly when not in the discovered list (no premature `model_not_found`).
- **no /metrics** — added GET `/api/metrics` + `/metrics` (stdlib Prometheus).
- **provider-session sync-state noise** — `cma_run_provider` no longer runs cross-account sync-state on isolated provider dirs.

Verified live (real "Do you see my code?" against production APIs)

- **9 providers verified:** DeepSeek, Groq, Mistral, Cerebras, Novita, NVIDIA, Cohere, Codestral, HuggingFace.
- **Both transports, both hosts:** native (DeepSeek) + router (Novita via ccr) on macOS and on nezha.
- Account-side failures (402/401/429/403) and non-OpenAI providers documented honestly; excluded under "valid users only" but kept fully supported.

Safety

- Provider dirs (`~/.claude-prov-*`) excluded from account detection — existing claudeN accounts and `claude-add-account` untouched.
- Secrets only in the keys file + on-host `mode-600 .env`; never in the repo. All published ports bound to loopback.

v1.0.0 — 2026-06-16 — Dynamic provider-alias generator

First tagged release. `claude-providers` creates per-provider Claude Code aliases (DeepSeek, Groq, GLM, ...) from your keys file pointed at each provider's strongest model — fully dynamic via `models.dev` + the `LLMsVerifier` submodule, hybrid native/claude-code-router transport, full lifecycle + tests + docs. See `docs/Provider_Aliases_User_Guide.md`.

v1.6.7 — 2026-06-21 — Poe proxy fix for all aliases

Fixed

- **Poe proxy not starting for poe2/poe3** — proxy detection used exact provider ID (poe2_proxy.py) which doesn't exist. Fixed to check base name too (poe_proxy.py for poe2, poe3 aliases).
- **lib.sh**: base proxy detection with `${CMA_PROVIDER_ID%%[0-9]*}`
- **alias file**: same fix applied

Verified

- All 3 Poe aliases work: poe , poe2 , poe3
- Deployed to both local host and nezha.local

Tests

- Local: 8/8 ALL GREEN
- nezha.local: 7/8 (pandoc missing — pre-existing)

v1.6.8 — 2026-06-21 — Poe proxy gzip fix

Fixed

- **Poe proxy gzip decompression** — Poe API returns gzip-compressed responses but the proxy tried to read them as UTF-8 without decompressing, causing UnicodeDecodeError. Added gzip decompression for both success and error responses.

Verified

- poe (claude-sonnet-4.6): YES
- poe2 (gpt-5.5): YES
- poe3 (grok-4): Different error (Grok-4 schema validation, not tools format)

Tests

- Local: 8/8 ALL GREEN
- nezha.local: 8/8 ALL GREEN

v1.6.9 — 2026-06-21 — Poe proxy \$ref fix for Grok-4

Fixed

- ****Poe proxy \$ref resolution**** — Claude Code sends tool schemas with ``$ref`references to$defs``. Grok-4 and some providers don't support ``$ref`in tool schemas`. Added `resolve_refs()` function that extracts `$defs``, resolves all ``$ref`references` to inline definitions, and removes `$defs``.

Verified

- poe (claude-sonnet-4.6): YES
- poe2 (gpt-5.5): YES
- poe3 (grok-4): YES (was failing, now works)

Tests

- Local: 8/8 ALL GREEN
- nezha.local: 8/8 ALL GREEN

v1.7.0 — 2026-06-22 — Poe proxy complete fix (all aliases verified)

Fixed

- **Poe proxy shared directory** — the proxy at `~/.local/share/.../proxy/poe_proxy.py` was the OLD version without gzip and \$ref fixes. `install.sh` copies from `scripts/` but the shared dir still had the old version. Fixed by ensuring updated proxy is copied to shared directory.
- **install.sh** now copies proxy scripts during installation (already in place)

Verified (all three aliases through full Claude Code flow)

- poe (claude-sonnet-4.6): YES
- poe2 (gpt-5.5): YES
- poe3 (grok-4): YES

Root Cause Analysis

The proxy had three issues:

1. **gzip** — Poe returns gzip-compressed responses, proxy didn't decompress
2. **\$ref** — Claude Code sends tool schemas with \$ref, Grok-4 doesn't support them
3. **shared dir** — Updated proxy wasn't copied to shared directory

All three fixed and verified.

Tests

- Local: 8/8 ALL GREEN
- nezha.local: 8/8 ALL GREEN

v1.7.1 — 2026-06-22 — Full validation + release

Fixed

- **Port-ready check** for proxy startup — replaced `sleep 1` with polling loop (`lsof -i`) ensuring proxy is listening before ccr config is written
- **Claude alias regression test** — 11 assertions proving claudeN aliases use `cma_run` (no proxy/transformer code), providers use `cma_run_provider`
- **Command injection fix** in `verify_aliases_live.sh` — replaced `bash -c` subshell with safe indirect expansion

Tests

- Local: **9/9 ALL GREEN** (new: `test_claude.sh` — 11 assertions)
- nezha.local: 8/9 (export fails — pandoc missing)

Release

- v1.7.1 — pushed to github, gitlab, gitflic, gitverse

v1.7.2 — 2026-06-22 — Claude alias verification, full release

Added

- **Claude alias verification** in `verify_aliases_live.sh` — tests claude1/2/3 alongside provider aliases
- **TOON tested** on all aliases — verified working

■

Tests

- Local: **9/9 ALL GREEN**
- nezha.local: 8/9 (pandoc missing)
- All claude1/2/3: OK
- All provider aliases: verified